

Module 5: Core Applications, Evaluation Metrics & Ethics in NLP

Topics 46 to 53 • Machine Translation BLEU, Summarization ROUGE, RAG, NER & Algorithmic De-Biasing

Module 5: Core Applications, Evaluation Metrics & Ethics in NLP

📌 Syllabus & Pedagogical Blueprint (Topics 46 to 53)

Text Classification & Sentiment Analysis • Aspect-Based Sentiment Analysis (ABSA) • Named Entity Recognition (NER) & Sequence Tagging Schemas (BIO, BIOES) • BiLSTM-CRF Architecture for NER • Machine Translation: Statistical (SMT) vs. Neural (NMT) • Beam Search Decoding Mechanics • Exact Mathematical Formulation of the BLEU Metric with Brevity Penalty • Text Summarization: Extractive (TextRank / LexRank) vs. Abstractive (Seq2Seq with Copy/Coverage Mechanisms) • ROUGE-1, ROUGE-2, ROUGE-L Evaluation • Question Answering (Extractive SQuAD exact match/F1 vs. Retrieval-Augmented Generation / RAG) • Dialogue Systems (Task-Oriented vs. Open-Domain LLM Chatbots) • Ethics, Algorithmic Bias, De-biasing Word Embeddings, LLM Hallucinations & Environmental Sustainability • 15 Solved University Examination Questions.

Topic 46: Machine Translation & The BLEU Evaluation Metric

The standard benchmark for automated machine translation evaluation is the **Bilingual Evaluation Understudy (BLEU – Papineni et al. 2002)** metric, measuring modified n-gram precision penalized by brevity:

🔍 Exact Mathematical Formulation of BLEU Score

$$\text{BLEU} = \text{BP} \cdot \exp \left(\sum_{n=1}^N w_n \ln p_n \right)$$

$$\text{Modified Precision: } p_n = \frac{\sum_{C \in \{\text{Candidates}\}} \sum_{\text{n-gram} \in C} \text{Count}_{\text{clip}}(\text{n-gram})}{\sum_{C \in \{\text{Candidates}\}} \sum_{\text{n-gram} \in C} \text{Count}(\text{n-gram})}$$

$$\text{Brevity Penalty (BP): } \text{BP} = \begin{cases} 1 & \text{if } c > r \\ \exp \left(1 - \frac{r}{c} \right) & \text{if } c \leq r \end{cases}$$

Where c is the candidate translation length and r is the effective reference corpus length (equal weights $w_n = 1/N$, typically $N = 4$).

🏠 Solved Numerical 5: Complete BLEU-1 and BLEU-2 Score Calculation

Candidate Translation ($c = 6$): "the cat the cat on the"

Reference Translation 1 ($r_1 = 6$): "the cat sat on the mat"

Reference Translation 2 ($r_2 = 6$): "there is a cat on mat"

1. Unigram Precision (p_1):

- Candidate unigrams: `the` (3 times), `cat` (2 times), `on` (1 time). Total candidate unigram count = 6.
- Max reference counts: `the` occurs max 2 times in Ref 1. `cat` occurs max 1 time. `on` occurs max 1 time.
- Clipped counts: $\text{Count}_{\text{clip}}(\text{the}) = 2$, $\text{Count}_{\text{clip}}(\text{cat}) = 1$, $\text{Count}_{\text{clip}}(\text{on}) = 1$.

$$p_1 = \frac{2 + 1 + 1}{6} = \frac{4}{6} = 0.6667$$

2. Bigram Precision (p_2):

- Candidate bigrams: `the cat` (2 times), `cat the` (1 time), `cat on` (1 time), `on the` (1 time). Total = 5 bigrams.
- Max reference counts: `the cat` (max 1), `cat the` (max 0), `cat on` (max 1 in Ref 2), `on the` (max 1 in Ref 1).
- Clipped counts: `the cat` (1), `cat the` (0), `cat on` (1), `on the` (1) $\implies \sum = 3$.

$$p_2 = \frac{3}{5} = 0.6000$$

3. Brevity Penalty ($c = 6, r = 6$): $c = r \implies \text{BP} = 1.0$.

$$\text{BLEU-2} = 1.0 \times \exp(0.5 \ln(0.6667) + 0.5 \ln(0.6000)) = \sqrt{0.6667 \times 0.6000} = \sqrt{0.4000} = 0.6325 \text{ (63.25\%)}$$

Topic 47: Text Summarization & The ROUGE Evaluation Metric

ROUGE (Recall-Oriented Understudy for Gisting Evaluation – Lin 2004) measures the n-gram recall of candidate summaries against gold-standard human reference summaries:

$$\text{ROUGE-N} = \frac{\sum_{s \in \{\text{References}\}} \sum_{n\text{-gram} \in s} \text{Count}_{\text{match}}(n\text{-gram})}{\sum_{s \in \{\text{References}\}} \sum_{n\text{-gram} \in s} \text{Count}(n\text{-gram})}$$

$$\text{ROUGE-L} = \frac{\text{LCS}(\text{Reference}, \text{Candidate})}{|\text{Reference}|} \quad (\text{Longest Common Subsequence})$$

Topic 48: Retrieval-Augmented Generation (RAG) Architecture

Large Language Models suffer from *hallucinations* and frozen knowledge cutoffs. **RAG (Lewis et al. 2020)** augments generative transformers with dynamic external retrieval:

1. **Dense Indexing**: Chunk knowledge base documents and encode chunks into dense vectors using a bi-encoder (e.g., DPR, BGE-embeddings). Index vectors into a vector database (FAISS, HNSW).
2. **Dense Retrieval**: Encode user query q and retrieve top- k most similar document passages via Cosine similarity.
3. **Prompt Synthesis & Generation**: Concatenate retrieved passages as context into the prompt: "Answer query Q strictly using context C " $\rightarrow$ feed into LLM generator.

Topic 49: Ethics, Algorithmic Bias & De-biasing Word Vectors

Word embeddings trained on uncensored corpora learn harmful societal and gender stereotypes (e.g., $\mathbf{v}_{\text{doctor}} - \mathbf{v}_{\text{man}} + \mathbf{v}_{\text{woman}} \approx \mathbf{v}_{\text{nurse}}$). Bolukbasi et al. (2016) proposed geometric de-biasing:

1. **Identify Gender Subspace (g)**: Compute PCA on difference vectors of defining gender pairs ($\mathbf{v}_{\text{she}} - \mathbf{v}_{\text{he}}, \mathbf{v}_{\text{woman}} - \mathbf{v}_{\text{man}}$).
2. **Hard De-biasing (Neutralization)**: Project gender-neutral words (e.g., `doctor`, `programmer`, `nurse`) to remove their component in the gender direction:

$$\mathbf{w}_{\text{neutralized}} = \mathbf{w} - (\mathbf{w} \cdot \mathbf{g})\mathbf{g}$$

3. **Equalization:** Ensure gender-neutral words have equal distance to paired gender words ($\|\mathbf{w} - \mathbf{v}_{\text{grandmother}}\| = \|\mathbf{w} - \mathbf{v}_{\text{grandfather}}\|$).

Topic 50: Master University Examination Question Bank (Module 5)

Q1. Compare Extractive Summarization (TextRank) vs. Abstractive Summarization (T5/BART). (8 Marks)

- **Extractive Summarization (TextRank):** Builds a sentence-level graph where edge weights represent sentence similarity ($w_{ij} = \frac{|S_i \cap S_j|}{\log |S_i| + \log |S_j|}$). Runs PageRank centrality algorithm to identify the top k most salient verbatim sentences. Guaranteed grammatically correct, but lacks cohesion and cannot paraphrase.
- **Abstractive Summarization:** Uses sequence-to-sequence transformers (BART, T5) to synthesize novel phrasing, compress clauses, and generate human-like summaries. Prone to hallucinating factual errors.

Q2. Explain Named Entity Recognition BIO vs BILOU Tagging Schemes. (6 Marks)

- **BIO (IOB2):** 'B-Type' (Beginning of entity), 'I-Type' (Inside continuation), 'O' (Outside). Example: '[New(B-LOC), York(I-LOC), City(I-LOC), is(O)]'.
- **BILOU:** Explicitly distinguishes singleton entities: 'B-Type' (Beginning), 'I-Type' (Inside), 'L-Type' (Last token of multi-token entity), 'O' (Outside), and 'U-Type' (Unique single-token entity). Gives higher discriminative power to CRF sequence labelers!

Topic 51: Beam Search Decoding Mechanics in Sequence Generation

Greedy search selects the single most probable next token $\arg\max_w P(w \mid y_{\leq t})$. Beam Search maintains a beam of the top B most probable candidate sequences:

🏛️ Solved Numerical 8: Step-by-Step Beam Search Trace ($B = 2$)

Translate input sentence with Beam Width $B = 2$ across 2 decoding time steps:

- **Step 1 ($t = 1$):** Vocabulary probabilities: $P("I") = 0.5$, $P("He") = 0.4$, $P("The") = 0.1$. $\text{Top 2 Active Beams: } \{\text{Beam}_1 = ["I"] \ (\text{score}=0.50), \text{Beam}_2 = ["He"] \ (\text{score}=0.40)\}$
- **Step 2 ($t = 2$):** Expand each of the 2 active beams:
 - From Beam_1 ("I"): $P("go" \mid "I") = 0.4 \implies 0.5 \times 0.4 = \mathbf{0.20}$; $P("run" \mid "I") = 0.3 \implies 0.5 \times 0.3 = \mathbf{0.15}$.
 - From Beam_2 ("He"): $P("goes" \mid "He") = 0.7 \implies 0.4 \times 0.7 = \mathbf{0.28}$; $P("runs" \mid "He") = 0.2 \implies 0.4 \times 0.2 = \mathbf{0.08}$.
- **Select Top $B = 2$ Candidates across all $2 \times |V|$ paths:**

Top Beam 1: ["He", "goes"] (score = **0.28**)

Top Beam 2: ["I", "go"] (score = **0.20**)

Length Normalization: Scores are normalized by $\frac{1}{T^\alpha}$ ($\alpha \approx 0.7$) to prevent unfair bias against longer sentence translations!

Topic 52: BiLSTM-CRF Architecture for Named Entity Recognition (NER)

Modern sequence labelers combine bidirectional recurrent neural feature extraction with global Conditional Random Field transition modeling:

1. **Word & Character Embeddings:** Words are represented via pre-trained embeddings concatenated with character-level CNN/BiLSTM vectors.
2. **BiLSTM Layer:** Forward LSTM ($\vec{\mathbf{h}}_t$) and backward LSTM ($\overleftarrow{\mathbf{h}}_t$) capture bidirectional unbounded contextual representations $\mathbf{h}_t = [\vec{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t]$.

3. **CRF Layer:** Learns a global state transition matrix $\mathbf{A}_{i,j} = P(\text{tag}_j \mid \text{tag}_i)$ that enforces valid syntactic constraints (e.g., prohibiting 'I-PER' immediately following 'O' without an intervening 'B-PER').

Topic 53: Advanced Downstream Tasks & University Exam Problem Bank (Part II)

Solved Numerical 13: ROUGE-1, ROUGE-2, and ROUGE-L Summarization Metrics Calculation

Generated Summary (C): "the prime minister visited london yesterday" (5 words)

Reference Summary (R): "prime minister visited london on monday" (6 words)

• **1. ROUGE-1 (Unigram Recall):**

- Reference unigrams: 'prime', 'minister', 'visited', 'london', 'on', 'monday' (Total = 6).
- Matching unigrams in candidate: 'prime', 'minister', 'visited', 'london' (4 matches).

$$\text{ROUGE-1 Recall} = \frac{4}{6} = \mathbf{0.6667 \text{ (66.67\%)}} \quad \text{Precision} = \frac{4}{5} = \mathbf{0.8000}$$

• **2. ROUGE-2 (Bigram Recall):**

- Reference bigrams: 'prime minister', 'minister visited', 'visited london', 'london on', 'on monday' (Total = 5).
- Matching bigrams: 'prime minister', 'minister visited', 'visited london' (3 matches).

$$\text{ROUGE-2 Recall} = \frac{3}{5} = \mathbf{0.6000 \text{ (60.00\%)}}$$

• **3. ROUGE-L (Longest Common Subsequence Recall):**

- $\text{LCS}(R, C) = [\text{"prime"}, \text{"minister"}, \text{"visited"}, \text{"london"}]$ (Length = 4).

$$\text{ROUGE-L Recall} = \frac{\text{Length}(\text{LCS})}{|\mathbf{R}|} = \frac{4}{6} = \mathbf{0.6667 \text{ (66.67\%)}}$$

Q3. Detail the Extractive Question Answering Architecture for SQuAD with Span Boundary Logits. (8 Marks)

In extractive Question Answering (e.g., SQuAD), input text is formatted as '[CLS] Query Tokens [SEP] Passage Context Tokens [SEP]'.

1. Passage contextual embeddings $\mathbf{h}_i \in \mathbb{R}^d$ are passed to two linear classification heads $\mathbf{w}_s, \mathbf{w}_e \in \mathbb{R}^d$.
2. Start and End span probabilities are computed via softmax across all context tokens:

$$P_{\text{start}}(i) = \frac{\exp(\mathbf{w}_s^T \mathbf{h}_i)}{\sum_j \exp(\mathbf{w}_s^T \mathbf{h}_j)} \quad P_{\text{end}}(j) = \frac{\exp(\mathbf{w}_e^T \mathbf{h}_j)}{\sum_k \exp(\mathbf{w}_e^T \mathbf{h}_k)}$$

3. The predicted answer span (i^*, j^*) maximizes $P_{\text{start}}(i) \times P_{\text{end}}(j)$ subject to constraint $i \leq j \leq i + L_{\text{max}}$.

Topic 53.3: Master University Exam Problem Bank (Part III)

Solved Problem 14: Aspect-Based Sentiment Analysis (ABSA) Pipeline

In ABSA, review sentence $S = \text{"The pasta was delicious but the service was terrible"}$ contains multiple polarities across distinct aspects:

1. **Aspect Term Extraction (ATE):** Sequence tagging identifies aspect terms: 'pasta' (Target 1), 'service' (Target 2).
2. **Aspect Term Polarity Classification (APTC):** Encodes sentence with target aspect marker: '[CLS] S [SEP] pasta [SEP] -> Positive (98%)'; '[CLS] S [SEP] service [SEP] -> Negative (95%)'.

Q4. Detail the Pointer-Generator Network for Abstractive Summarization with Out-Of-Vocabulary Copying. (8 Marks)

Pointer-Generator Networks (See et al. 2017) solve two major limitations of standard seq2seq summarizers: factual hallucinations and inability to produce out-of-vocabulary names/numbers.

1. **Generation Probability** $p_{\text{gen}} \in [0, 1]$: Computes a soft gating switch from decoder context $\mathbf{c}_t$, hidden state $\mathbf{s}_t$, and input $\mathbf{x}_t$:

$$p_{\text{gen}} = \sigma(\mathbf{w}_c^T \mathbf{c}_t + \mathbf{w}_s^T \mathbf{s}_t + \mathbf{w}_x^T \mathbf{x}_t + b_{\text{ptr}})$$

2. **Hybrid Output Distribution**: Interpolates between generating from vocabulary and copying directly from source text via attention weights a_i^t :

$$\mathbf{P}(\mathbf{w}) = p_{\text{gen}} \mathbf{P}_{\text{vocab}}(\mathbf{w}) + (1 - p_{\text{gen}}) \sum_{\mathbf{i}: \mathbf{w}_i = \mathbf{w}} \mathbf{a}_i^t$$

3. **Coverage Mechanism**: Maintains a running sum of attention weights $c_i^t = \sum_{t'=1}^{t-1} a_i^{t'}$ and penalizes repeatedly attending to the same source words, completely eliminating repetitive loop summaries!

Q5. Explain Statistical Watermarking for LLM-Generated Text (Kirchenbauer et al. 2023). (8 Marks)

To detect AI-generated text without changing perplexity:

1. When sampling token t , use previous token hash $h(x_{t-1})$ as a pseudo-random seed to partition the vocabulary V into a **Green List** G (size $\gamma|V|$) and a **Red List** R .

2. Add a positive bias δ to the logits of green tokens: $\tilde{z}_k = z_k + \delta$ for $k \in G$.

3. **Statistical Detection Test**: For a candidate text of T tokens, count green tokens $|T|_G$. Under human text, $|T|_G \sim \text{Binomial}(T, \gamma)$. Under watermarked text, $|T|_G \gg \gamma T$. A one-sided Z -score $z = \frac{|T|_G - \gamma T}{\sqrt{T\gamma(1-\gamma)}} > 4.0$ detects synthetic AI text with $p < 0.0001$ error rate!

🏛️ Solved Problem 19: Dense Passage Retrieval (DPR) Dual-Encoder Dot-Product Scoring

In modern Retrieval-Augmented Generation (RAG), traditional BM25 lexical search is replaced with dense semantic retrieval:

- **Question Encoder (E_Q)**: Maps user query q to vector $\mathbf{q} = E_Q(q) \in \mathbb{R}^{768}$.
- **Passage Encoder (E_P)**: Maps passage text p to vector $\mathbf{p} = E_P(p) \in \mathbb{R}^{768}$.
- **Retrieval Relevance Score**: $\text{sim}(q, p) = \mathbf{q}^T \mathbf{p}$.
- Trained via Negative Log-Likelihood loss with in-batch negatives:

$$\mathcal{L}(\mathbf{q}, \mathbf{p}^+, \mathbf{p}_1^-, \dots, \mathbf{p}_n^-) = -\log \frac{\exp(\mathbf{q}^T \mathbf{p}^+)}{\exp(\mathbf{q}^T \mathbf{p}^+) + \sum_{j=1}^n \exp(\mathbf{q}^T \mathbf{p}_j^-)}$$

Q6. Explain the Carbon Footprint & Environmental Impact of Pretraining Large Language Models. (6 Marks)

Pretraining frontier LLMs (GPT-4, LLaMA-3 405B) requires clusters of thousands of high-power GPUs (Nvidia H100) running continuously for months, consuming megawatts of electrical power (hundreds of megawatt-hours) and generating tons of equivalent CO_2 carbon emissions.

• **Mitigation Strategies**: (1) Energy-efficient data center siting near renewable hydro/solar power, (2) Parameter-Efficient Fine-Tuning (LoRA / QLoRA), (3) Model quantization (4-bit GPTQ, AWQ, GGUF), and (4) Knowledge distillation into compact edge SLMs (Phi-3, Gemma).

🏛️ Solved Problem 23: Complete METEOR & chrF Translation Evaluation Metrics

- **METEOR (Banerjee & Lavie 2005)**: Computes unigram harmonic mean precision and recall ($F_{\text{mean}} = \frac{10PR}{R+9P}$) with explicit penalty for fragmented word order ($\text{Penalty} = 0.5 \times (\frac{\# \text{chunks}}{\# \text{unigrams}})^3$), incorporating exact match, WordNet synsets, and Porter stems.
- **chrF (Popović 2015)**: Measures character n-gram F-score (typically 6-grams), completely eliminating language-dependent tokenizers and achieving superior correlation with human judgments on morphologically complex languages!

🏛️ Solved Problem 24: Dialogue State Tracking (DST) & Belief State Representation

In Task-Oriented Dialogue systems (MultiWOZ benchmark), the **Belief State** tracks user slot-value pairs across turns:

Belief State: {domain: 'restaurant', food: 'italian', price: 'moderate', area: 'centre', people: 4}

Dialogue Policy $\pi(a_t | s_t)$ maps belief state s_t to system action $a_t = \text{Offer}(\text{Restaurant_Name} = \text{"Pizza Express"})$, and NLG converts action to natural utterance.

🏛️ Solved Problem 34: TextRank Sentence Graph PageRank Convergence Formula

In TextRank (Mihalcea & Tarau 2004), let $G = (V, E)$ be a graph where nodes are sentences and edge weights w_{ij} represent token overlap similarity:

$$\mathbf{WS}(V_i) = (1 - d) + d \sum_{V_j \in \text{In}(V_i)} \frac{w_{ji}}{\sum_{V_k \in \text{Out}(V_j)} w_{jk}} \mathbf{WS}(V_j) \quad (d = 0.85)$$

Iterates power method until $|\mathbf{WS}^{(t+1)} - \mathbf{WS}^{(t)}| < 10^{-4}$, extracting top k sentences with highest stationary centrality scores!

🏛️ Solved Problem 35: Contrastive Representation Learning with InfoNCE Loss

In modern dense retrieval (DPR, Contriever), embeddings are optimized using the InfoNCE objective with temperature $\tau = 0.05$:

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(\text{sim}(\mathbf{q}, \mathbf{d}^+)/\tau)}{\exp(\text{sim}(\mathbf{q}, \mathbf{d}^+)/\tau) + \sum_{j=1}^K \exp(\text{sim}(\mathbf{q}, \mathbf{d}_j^-)/\tau)}$$

🏛️ Solved Problem 36: Quantization of Large Language Models (GPTQ vs AWQ)

- **GPTQ (Frantar et al. 2022)**: Second-order Taylor approximation of layer weights minimizing inverse Hessian reconstruction error H^{-1} .
- **AWQ (Activation-aware Weight Quantization - Lin et al. 2023)**: Protects the top 1% salient weight channels that correspond to large activation magnitudes, quantizing the remaining 99% weights to 4-bit integers with zero perplexity degradation!

🏛️ Solved Problem 43: COMET & Neural Translation Evaluation Metrics

Traditional n-gram metrics (BLEU, ROUGE) penalize valid translations that use synonyms (e.g., 'automobile' vs 'car'). **COMET (Rei et al. 2020)** uses cross-lingual pretrained encoders (XLM-RoBERTa) to project the source sentence, candidate translation, and reference into a shared multilingual embedding space, computing neural semantic similarity that achieves state-of-the-art correlation with human professional translators!

🏛️ Solved Problem 44: Direct Preference Optimization (DPO) Closed-Form Loss Derivation

Rafailov et al. (2023) proved that the optimal policy under Bradley-Terry reward models satisfies: $r(x, y) = \beta \log \frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)} + \beta \log Z(x)$.

$$\mathcal{L}_{\text{DPO}}(\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(\mathbf{x}, \mathbf{y}_w, \mathbf{y}_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(\mathbf{y}_w | \mathbf{x})}{\pi_{\text{ref}}(\mathbf{y}_w | \mathbf{x})} - \beta \log \frac{\pi_\theta(\mathbf{y}_l | \mathbf{x})}{\pi_{\text{ref}}(\mathbf{y}_l | \mathbf{x})} \right) \right]$$

Eliminates the separate reward model and reinforcement learning training stability issues entirely!

🏛️ Solved Problem 52: BLEU vs NIST Metric Comparison

While BLEU gives equal weight to all n -grams ($w_n = 1/4$), the **NIST Metric** weights each matching n -gram by its Information Content: $\text{Info}(w_1 \dots w_n) = \log_2 \frac{C(w_1 \dots w_{n-1})}{C(w_1 \dots w_n)}$, heavily rewarding rare informative content words over common stopwords!

🏛️ Solved Problem 53: Factuality Evaluation in Abstractive Summarization (QAGS & FactCC)

- **QAGS (Wang et al. 2020)**: Automatically generates QA pairs from the summary and verifies whether the source document answers them identically.
- **FactCC (Kryscinski et al. 2020)**: Uses a binary classification BERT model trained on syntactically mutated sentences to detect hallucinations!

🏛️ Solved Problem 60: BLEU Brevity Penalty Exponential Decay Curve

For candidate length $c = 10$ and reference length $r = 15$: $\text{BP} = \exp(1 - 15/10) = \exp(-0.5) \approx \mathbf{0.6065}$. The score is penalized by 40% for omitting essential reference content!

🏛️ Solved Problem 61: RAG Dense Indexing with FAISS Flat vs HNSW Graphs

- **IndexFlatIP**: Exact brute-force inner-product search $O(N \cdot d)$.
- **HNSW (Hierarchical Navigable Small World)**: Multi-layer proximity graph achieving approximate nearest neighbor retrieval in blazing fast $O(\log N)$ logarithmic time!

🏛️ Solved Problem 68: Complete 4-Gram BLEU Score Calculation Trace

Given candidate $c = 12$, reference $r = 14$, clipped precisions: $p_1 = 0.833, p_2 = 0.636, p_3 = 0.500, p_4 = 0.333$:

$$\text{BP} = \exp(1 - 14/12) = \exp(-0.1667) = \mathbf{0.8465}$$

$$\text{BLEU-4} = \mathbf{0.8465} \times \exp \left(\frac{\ln(\mathbf{0.833}) + \ln(\mathbf{0.636}) + \ln(\mathbf{0.500}) + \ln(\mathbf{0.333})}{4} \right) = \mathbf{0.8465} \times \mathbf{0.545} = \mathbf{0.4613} \text{ (46.13\%)}$$

🏛️ Solved Problem 69: ROUGE-L Longest Common Subsequence Dynamic Programming Recurrence

For reference $R[1..m]$ and candidate $C[1..n]$:

$$\text{LCS}[i, j] = \begin{cases} \text{LCS}[i-1, j-1] + 1 & \text{if } R[i] = C[j] \\ \max(\text{LCS}[i-1, j], \text{LCS}[i, j-1]) & \text{if } R[i] \neq C[j] \end{cases}$$

🏛️ Solved Problem 73: Text Summarization Abstractive Length Penalty Proof

In Beam Search for abstractive summarization, the length normalization penalty $\text{lp}(Y) = \frac{(5+|Y|)^\alpha}{(5+1)^\alpha}$ ($\alpha = 0.6$) prevents the decoder from prematurely outputting the ``end-of-sequence token!

🏛️ Solved Problem 77: Cross-Lingual Semantic Retrieval via Multilingual Sentence-BERT

In cross-lingual RAG, Sentence-BERT projects multilingual query q_{Hindi} and English document passage p_{English} into a unified shared semantic sphere, achieving high cosine similarity $\cos(\mathbf{q}, \mathbf{p}) > 0.85$ without explicit translation!

🏛️ Solved Problem 83: Exact Match (EM) vs Token-Level F1 in SQuAD QA

For prediction "the golden gate bridge" and gold answer "golden gate bridge":

- **Exact Match (EM):** Binary 0.0 (Strict string inequality).
- **Token-Level Precision:** $3/4 = 0.75$; **Recall:** $3/3 = 1.0$ $\implies \text{F1} = \frac{2(0.75)(1.0)}{0.75+1.0} = 0.8571$ (85.71%).

🏛️ Solved Problem 84: Algorithmic Debiasing Hard-Neutralization Matrix Projection

For gender direction $\mathbf{g} = (0.6, 0.8)^T$ and biased word vector $\mathbf{w}_{\text{doctor}} = (2.0, 1.0)^T$:

$$\mathbf{w} \cdot \mathbf{g} = 2.0(0.6) + 1.0(0.8) = 1.2 + 0.8 = 2.0$$

$$\mathbf{w}_{\text{neutral}} = (2.0, 1.0)^T - 2.0(0.6, 0.8)^T = (2.0 - 1.2, 1.0 - 1.6)^T = (0.8, -0.6)^T$$

Verification: $\mathbf{w}_{\text{neutral}} \cdot \mathbf{g} = 0.8(0.6) + (-0.6)(0.8) = 0.48 - 0.48 = 0.000$ (Perfect Orthogonality!)

🏛️ Solved Problem 89: Neural Machine Translation Beam Search Length Penalty Graph

Plot the cumulative log-probability $\sum_{t=1}^T \log P(y_t \mid y_{<t})$

🏛️ Solved Problem 90: RAG Vector Database Indexing with Inverted File Quantization (IVF-PQ)

IVF-PQ partitions 100 million document embeddings into 4096 Voronoi cell clusters and quantizes 768D float vectors into 64 bytes using Product Quantization, enabling billion-scale sub-millisecond retrieval!

🏛️ Solved Problem 95: Cross-Encoder vs Bi-Encoder RAG Reranking Pipeline

- **Bi-Encoder (Fast Retrieval):** Projects query and documents into isolated vectors; searches top 100 documents via FAISS in 5ms.
- **Cross-Encoder (High Precision Reranker):** Concatenates query and document `[CLS] Query [SEP] Doc [SEP]` through full cross-attention layers, sorting the top 5 most relevant passages with +**12.4% MRR@10** accuracy boost!

🏛️ Solved Problem 96: Watermark Detection Z-Score Calculation

For text of $T = 200$ tokens with $\gamma = 0.5$, an AI detector observes $|T|_G = 160$ green tokens:

$$\mu = 200(0.5) = 100 \quad \sigma = \sqrt{200(0.5)(0.5)} = \sqrt{50} \approx 7.071$$

$$z = \frac{160 - 100}{7.071} = \frac{60}{7.071} = +8.485 \implies p < 10^{-16} \quad (\text{Definitive Proof of Synthetic AI Origin!})$$

🏛️ Solved Problem 100: BLEU Brevity Penalty Strict Monotonicity Proof

Prove that $\frac{d}{dc} \text{BP}(c, r) > 0$ for all candidate lengths $c < r$, showing that adding correct words strictly monotonically increases the penalty multiplier!

🏛️ Solved Problem 104: TextRank Sentence Centrality Power Method Trace

Starting from uniform initial scores $WS_0 = (1/N, \dots, 1/N)^T$, power iteration multiplies by Google matrix $\mathbf{M} = d\mathbf{T} + \frac{1-d}{N}\mathbf{E}$, converging in < 20 matrix-vector multiplications!

🏛️ Solved Problem 107: Beam Search Decoding Node Expansion Budget

With beam width $B = 4$ and max sequence length $T = 50$, search space explores $B \times |V| = 4 \times 32000 = 128,000$ hypotheses per step, retaining only top 4 survivors!

🏛️ Solved Problem 112: ROUGE-2 Precision vs Recall Tradeoff in Document Summaries

Recall measures factual completeness ($\frac{\text{matching bigrams}}{\text{reference bigrams}}$), while Precision measures conciseness ($\frac{\text{matching bigrams}}{\text{summary bigrams}}$). F_1 -harmonic mean provides balanced evaluation!

🏛️ Solved Problem 113: SQuAD 2.0 Unanswerable Question Classification Threshold

SQuAD 2.0 incorporates unanswerable questions. The model outputs no-answer probability $P(\text{no-answer}) = \text{softmax}(\mathbf{w}_{\text{null}}^T \mathbf{h}_{[\text{CLS}]})$. If $P > \tau$ (threshold $\tau \approx 0.5$), the system abstains from generating a span!

🏛️ Solved Problem 118: RAG Hybrid Search Reciprocal Rank Fusion (RRF) Trace

Given BM25 rank $r_{\text{BM25}}(d) = 2$ and Dense Vector rank $r_{\text{Dense}}(d) = 5$ with constant $k = 60$:

$$\text{RRF}(d) = \frac{1}{60 + 2} + \frac{1}{60 + 5} = \frac{1}{62} + \frac{1}{65} = 0.01613 + 0.01538 = 0.03151$$

Combines exact lexical precision with semantic generalization without score normalization!

🏛️ Solved Problem 119: Generative Text Watermark Green-List Generation Trace

Given preceding token hash $h = 42$, pseudo-randomly partition vocabulary V ($|V| = 32,000$) into green list G (16,000 tokens) with bias $\delta = 2.0$, forcing $> 80\%$ of sampled tokens into G !

Solved Problem 121: Translation Evaluation Metric Selection Guidelines

- **BLEU**: Standard academic baseline (fast n-gram exact match).
- **ROUGE**: Standard for summarization (recall-focused).
- **COMET**: Neural metric with highest correlation to human translation quality across 100+ languages!